

InkBrush: A Flexible and Controllable Authoring Tool for 3D Ink Painting

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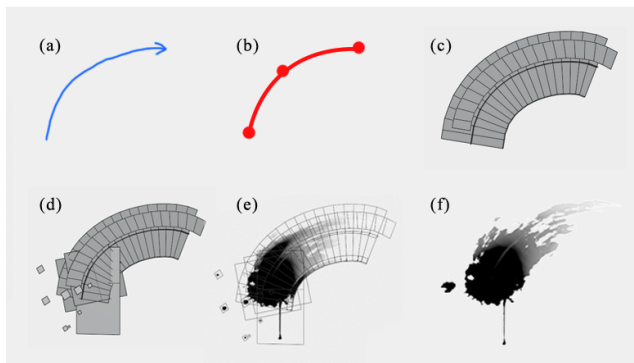


Figure 1: The process of generating strokes in Inkbrush (left) and 3D ink cat authored with Inkbrush (right).

ABSTRACT

As a stylized representation of traditional painting art, ink painting has been widely used in most of Asian countries. Art creations with ink painting strokes in 3D space have shown potential in animation and games. We propose a sketch-based authoring tool for drawing ink painting style strokes in 3D space. With automatic strip-based modeling and procedural texture generation, users can easily create ink painting style artworks by drawing sketches. Our system generates ink effects in real time and provides a simple non-linear workflow. User evaluations show that novice users can easily use the system and quickly create 3D ink paintings.

CCS CONCEPTS

• Computing methodologies → Parametric curve and surface models.

KEYWORDS

Ink painting, sketching, procedural modeling, authoring

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1 INTRODUCTION

Ink painting is a popular traditional art form in Asian countries, dating back as far as the Han Dynasty in China. With the increasing demand for 3D digital materials in industries such as games, animation and VR, 3D ink painting has become popular as a novel form[2, 5]. Compared to conventional 3D works, creating aesthetically pleasing 3D ink paintings is quite challenging, mainly because of the difficulty of rendering diverse ink strokes in 3D space.

Recently, there have been various developments in 3D ink painting. Some creators use software like 3ds Max and Maya to directly draw ink textures on objects. However, these tools are primarily geared towards experienced artists, making them difficult for novices to learn. Other techniques like VR painting allow for drawing from different angles but can be burdensome and result in messy strokes. Non-photorealistic rendering (NPR) methods provide better results from any angle but require high computing power and limit artist involvement in the creative process[1, 3, 6, 7].

In this paper, we propose Inkbrush, a novel sketch-based creation tool that allows artists to create 3D ink painting artwork conveniently using digital brushes. Users only need to draw freely, and

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the system will automatically produce detailed ink strokes to go along with each drawing input. With minimum modification of users' input and stylized ink stroke rendering, Inkbrush provides an intuitive and high quality authoring experience for users.

The major contributions of this work are summarized as follows:

- We present an intuitive and user-friendly interface for creating 3D ink paintings, maintaining a harmonious balance between the creative process and computer generation.
- We propose a real-time and scalable method for generating 3D ink painting stroke models, suitable for applications in animation and gaming.
- We explore the design method of converting 2D ink paintings to 3D space, providing a versatile approach applicable to other painting styles' 3D transformation.

2 INKBRUSH

Inkbrush is a 3D model based brush that can be viewed from various angles (shown in Figure 2). Our system first preprocesses each sketch input by the user, and then generates the main model using the processed sketch and user input parameters. Next, based on the main mesh, we automatically generate adjustable ink textures and various ink details such as ink dots and ink drops (shown in Figure 1 left). Below we will introduce the main components of the brush, followed by the user interface.



Figure 2: Left to right: unrendered model, front view, side view, top view

2.1 Main mesh

The main mesh is the primary part of each stroke, which determines the direction and shape of each stroke and is the basis for further model generation. Each main mesh consists of a central core mesh and accessory meshes. The core mesh is generated directly from the sketch drawn by the user, and the accessory meshes are generated by random offset based on the geometry of the core mesh.

2.2 Procedural ink texture

Ink textures are the most dominant visual element of brush strokes, which are two-dimensional ink images with transparent channels generated based on the geometric information of the main mesh. Each texture is generated by a base map and a mask. The mask determines the outer contours and edge details of the brush strokes, while the base map describes the internal texture features. Different textures and outlines can be combined to form different visual effects. Outlines are mainly used to simulate the amount of ink, while textures describe textural properties of the rice paper.

We summarized five important dimensions: humidity, blackness, half-dry, trailing, and color (shown in Figure 3). These five dimensions are independent of each other, and the artist is free to adjust.

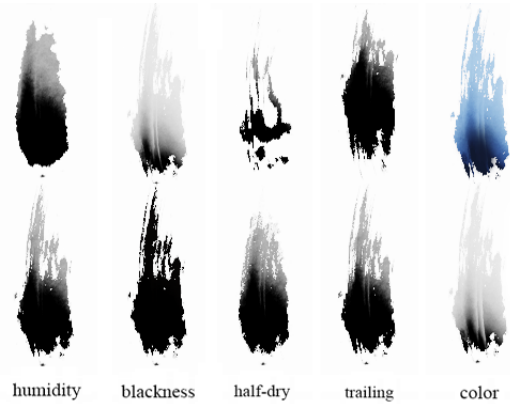


Figure 3: The differences in humidity, blackness, half-dry, trailing and color

2.3 Drops and Dots

The ink dots and drops are quads with textures, which greatly improves the performance and allows our system to host a large number of ink dots (shown in Figure 1 left). In particular, to ensure that they do not disappear at certain angles, we put a rotation constraint on them, and they will always be oriented towards the viewing point. Dots and drops have multi-dimensional parameters such as quantity, position, and size that can be adjusted.

2.4 Animation

The stroke animation is generated fully automatically, as the drawing sequence is recorded during the artist's drawing process. The user can play back the animation by adjusting the parameters after finishing a section of drawing (Figure 4).



Figure 4: "UIST" was written using Inkbrush and played the animation with parameters (0-100).

2.5 User interface

To make the system easy to use, we took inspiration from the traditional drawing process and designed a non-linear workflow that consists of two parts: Drawing and Inking. When users are authoring, they can switch back and forth between these two parts at any time. In our user interface, the 3D view canvas on the left is the drawing workspace, and the vertical bar on the right is the

inking workspace (Figure 5). The painting tools are placed at the left side of the drawing workspace, while the inking workspace contains mainly various parameters.

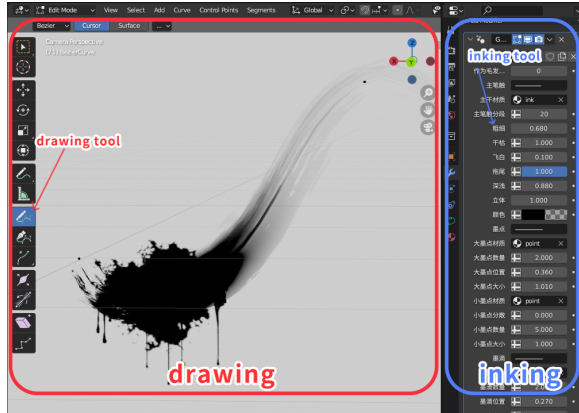


Figure 5: The user interface,

3 USER STUDY

We released Inkbrush worldwide and have over 20,000 downloads and collected some user feedback and user works (Figure 6). User evaluations show Inkbrush is easy to use and novice users can quickly create satisfying ink works. Professional Chinese painters evaluate that this brush is able to express the style and beauty of Chinese painting in three dimensions.

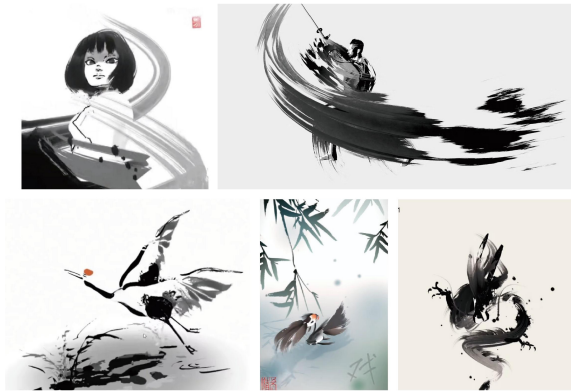


Figure 6: Works from some of our users.

4 LIMITATION AND FUTURE WORK

Although we offer a novel approach to creating 3D ink paintings, Inkbrush is not good at drawing objects that are mainly blocks. We consider adding a sketch modeling approach similar to Simpm modeling[4] to make the 3D ink painting process complete.

Since Inkbrush uses a strip-based modeling approach, they tend to appear unnaturally at the intersections of different meshes as well as at their own intersections. We considered solutions in two

directions, one is to possibly post-process based on screen space, blurring on screen space for the problems inside the brush; the other way is to align the UVs of different meshes exactly at the seams to ensure strict consistency of textures and thus eliminate the unnatural seams.

5 CONCLUSION

In this paper, we present Inkbrush, a 3D ink brush based on sketch modeling, which allows users to draw simple sketches to achieve the artistic effect of ink painting. Inkbrush is able to generate effects in real time during the creation process, providing a highly controllable and straightforward workflow. Our system strikes a good balance between creative intent and automatic generation, ensuring creative freedom. User evaluations show that even beginners with no experience in 3D design can quickly create intriguing 3D ink paintings. The implementation method is universal and provides ideas and references for the process of migrating 2D brushes to 3D brushes.

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